

Design Rules Verification Report

Filename : C:\Users\Public\Documents\Altium\Manea_TCAD_PROIECT\Manea_ThCAD_F

Warnings 8
Rule Violations 137

Warnings	
Unplated multi-layer pad(s) detected	8
Total	8

Rule Violations	
Clearance Constraint (Gap=0.254mm) (All),(All)	0
Short-Circuit Constraint (Allowed=No) (All),(All)	0
Un-Routed Net Constraint (All)	8
Modified Polygon (Allow modified: No), (Allow shelved: No)	0
Width Constraint (Min=0.254mm) (Max=0.254mm) (Preferred=0.254mm) (All)	0
Width Constraint (Min=0.7mm) (Max=1.5mm) (Preferred=0.8mm) (InNet('12V 1A 1'))	0
Width Constraint (Min=0.7mm) (Max=1.5mm) (Preferred=0.8mm) (InNet('12V 1A 2'))	0
Width Constraint (Min=0.2mm) (Max=0.25mm) (Preferred=0.25mm) (InxNetClass('All xNets'))	0
Width Constraint (Min=0.2mm) (Max=0.5mm) (Preferred=0.5mm) (InNetClass('NewClass'))	0
Routing Topology Rule(Topology=Shortest) (All)	0
Routing Layers(All)	0
Routing Via (MinHoleWidth=0.6mm) (MaxHoleWidth=0.6mm) (PreferredHoleWidth=0.6mm) (MinWidth=1mm,	0
Differential Pairs Uncoupled Length using the Gap Constraints (Min=0.254mm) (Max=0.254mm) (Prefered=0.254mm)	0
Power Plane Connect Rule(Relief Connect)(Expansion=0.508mm) (Conductor Width=0.254mm) (Air Gap=0.254mm,	0
Fabrication Testpoint Style (Under Component=Yes) (All)	0
Fabrication Testpoint Usage (Valid =One Required, Allow multiple per net=No) (All)	35
Assembly Testpoint Style (Under Component=Yes) (All)	0
Assembly Testpoint Usage (Valid =One Required, Allow multiple per net=No) (All)	35
Hole Size Constraint (Min=0.025mm) (Max=2.54mm) (All)	16
Pads and Vias to follow the Drill pairs settings	0
Hole To Hole Clearance (Gap=0.254mm) (All),(All)	0
Minimum Solder Mask Sliver (Gap=0.254mm) (All),(All)	6
Silk To Solder Mask (Clearance=0.254mm) (IsPad),(All)	36
Silk to Silk (Clearance=0.254mm) (All),(All)	1
Net Antennae (Tolerance=0mm) (All)	0
Component Clearance Constraint (Horizontal Gap = 0.254mm, Vertical Gap = 0.254mm) (All),(All)	0
Height Constraint (Min=0mm) (Max=25.4mm) (Prefered=12.7mm) (All)	0
Supply Net(s)(InNet('GND'))	0
Total	137

Unplated multi-layer pad(s) detected	
Pad SPK3-1(177.4mm,117mm) on Multi-Layer on Net GND	
Pad SPK2-1(217.9mm,47.5mm) on Multi-Layer on Net GND	
Pad SPK4-1(217.9mm,89mm) on Multi-Layer on Net GND	
Pad SPK1-1(176.4mm,22.5mm) on Multi-Layer on Net GND	
Pad SPK1-2(151.6mm,22.5mm) on Multi-Layer on Net NetC9_2	
Pad SPK4-2(193.1mm,89mm) on Multi-Layer on Net NetC18_2	
Pad SPK2-2(193.1mm,47.5mm) on Multi-Layer on Net NetC10_2	
Pad SPK3-2(152.6mm,117mm) on Multi-Layer on Net NetC19_2	

Un-Routed Net Constraint (All)
Un-Routed Net Constraint: Net NetC19_2 Between Pad C19-2(126.45mm,123.7mm) on Top Layer And Pad SPK3-2(152.6mm,117mm) on Multi-Layer
Un-Routed Net Constraint: Net GND Between Pad SPK1-1(176.4mm,22.5mm) on Multi-Layer [Unplated] And Track
Un-Routed Net Constraint: Unplated Pad SPK1-2(151.6mm,22.5mm) on Multi-Layer
Un-Routed Net Constraint: Net GND Between Track (217.9mm,47.5mm)(217.9mm,47.5mm) on Top Layer And Pad SPK2-1(217.9mm,47.5mm) or
Un-Routed Net Constraint: Net NetC10_2 Between Track (186.971mm,46.971mm)(192.571mm,46.971mm) on Top Layer And Pad
Un-Routed Net Constraint: Net GND Between Pad SPK3-1(177.4mm,117mm) on Multi-Layer [Unplated] And Via (177.5mm,98mm) from Top Layer to
Un-Routed Net Constraint: Net GND Between Pad SPK4-1(217.9mm,89mm) on Multi-Layer [Unplated] And Track (217.9mm,89mm)(217.9mm,106.9mm) or
Un-Routed Net Constraint: Unplated Pad SPK4-2(193.1mm,89mm) on Multi-Layer

Fabrication Testpoint Usage (Valid =One Required, Allow multiple per net=No) (All)
Fabrication Testpoint Usage: Net 12V 1A 1 - Missing required testpoint
Fabrication Testpoint Usage: Net 12V 1A 2 - Missing required testpoint
Fabrication Testpoint Usage: Net GND - Missing required testpoint
Fabrication Testpoint Usage: Net NetC1_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC10_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC10_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC11_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC12_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC12_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC13_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC14_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC14_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC15_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC16_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC17_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC18_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC18_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC19_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC19_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC2_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC20_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC3_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC3_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC4_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC5_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC5_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC6_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC7_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC8_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC9_1 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC9_2 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC2_7 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC2_8 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC3_7 - Missing required testpoint
Fabrication Testpoint Usage: Net NetC3_8 - Missing required testpoint

Assembly Testpoint Usage (Valid =One Required, Allow multiple per net=No) (All)
Assembly Testpoint Usage: Net 12V 1A 1 - Missing required testpoint
Assembly Testpoint Usage: Net 12V 1A 2 - Missing required testpoint
Assembly Testpoint Usage: Net GND - Missing required testpoint
Assembly Testpoint Usage: Net NetC1_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC10_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC10_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC11_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC12_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC12_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC13_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC14_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC14_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC15_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC16_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC17_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC18_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC18_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC19_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC19_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC2_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC20_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC3_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC3_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC4_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC5_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC5_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC6_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC7_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC8_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC9_1 - Missing required testpoint
Assembly Testpoint Usage: Net NetC9_2 - Missing required testpoint
Assembly Testpoint Usage: Net NetC2_7 - Missing required testpoint
Assembly Testpoint Usage: Net NetC2_8 - Missing required testpoint
Assembly Testpoint Usage: Net NetC3_7 - Missing required testpoint
Assembly Testpoint Usage: Net NetC3_8 - Missing required testpoint

Hole Size Constraint (Min=0.025mm) (Max=2.54mm) (All)
Hole Size Constraint (9.3mm > 2.54mm) Pad D1-D4-1(21.25mm,51.75mm) on Multi-Layer Actual Slot Hole Width = 9.3mm
Hole Size Constraint (9.3mm > 2.54mm) Pad D1-D4-2(36.75mm,57.75mm) on Multi-Layer Actual Slot Hole Width = 9.3mm
Hole Size Constraint (9.3mm > 2.54mm) Pad D1-D4-3(15.25mm,67.25mm) on Multi-Layer Actual Slot Hole Width = 9.3mm
Hole Size Constraint (9.3mm > 2.54mm) Pad D1-D4-4(30.75mm,73.25mm) on Multi-Layer Actual Slot Hole Width = 9.3mm
Hole Size Constraint (3mm > 2.54mm) Pad Free-15(225mm,135.5mm) on Multi-Layer Actual Hole Size = 3mm
Hole Size Constraint (3mm > 2.54mm) Pad Free-15(225mm,5mm) on Multi-Layer Actual Hole Size = 3mm
Hole Size Constraint (3mm > 2.54mm) Pad Free-15(4.5mm,5mm) on Multi-Layer Actual Hole Size = 3mm
Hole Size Constraint (3mm > 2.54mm) Pad Free-15(5mm,135.5mm) on Multi-Layer Actual Hole Size = 3mm
Hole Size Constraint (4.4mm > 2.54mm) Pad SPK1-1(176.4mm,22.5mm) on Multi-Layer Actual Hole Size = 4.4mm
Hole Size Constraint (4.4mm > 2.54mm) Pad SPK1-2(151.6mm,22.5mm) on Multi-Layer Actual Hole Size = 4.4mm
Hole Size Constraint (4.4mm > 2.54mm) Pad SPK2-1(217.9mm,47.5mm) on Multi-Layer Actual Hole Size = 4.4mm
Hole Size Constraint (4.4mm > 2.54mm) Pad SPK2-2(193.1mm,47.5mm) on Multi-Layer Actual Hole Size = 4.4mm
Hole Size Constraint (4.4mm > 2.54mm) Pad SPK3-1(177.4mm,117mm) on Multi-Layer Actual Hole Size = 4.4mm
Hole Size Constraint (4.4mm > 2.54mm) Pad SPK3-2(152.6mm,117mm) on Multi-Layer Actual Hole Size = 4.4mm
Hole Size Constraint (4.4mm > 2.54mm) Pad SPK4-1(217.9mm,89mm) on Multi-Layer Actual Hole Size = 4.4mm
Hole Size Constraint (4.4mm > 2.54mm) Pad SPK4-2(193.1mm,89mm) on Multi-Layer Actual Hole Size = 4.4mm

Minimum Solder Mask Sliver (Gap=0.254mm) (All), (All)
Minimum Solder Mask Sliver Constraint (0.156mm < 0.254mm) Between Pad IC3-6(68.88mm,21.96mm) on Multi-Layer And Via (67.2mm,22.2mm) from
Minimum Solder Mask Sliver Constraint (0.047mm < 0.254mm) Between Pad R1-1(21.575mm,27.5mm) on Top Layer And Pad R1-2(22.425mm,27.5mm)
Minimum Solder Mask Sliver Constraint (0.047mm < 0.254mm) Between Pad R12-1(24.075mm,110mm) on Top Layer And Pad R12-2(24.925mm,110mm)
Minimum Solder Mask Sliver Constraint (0.047mm < 0.254mm) Between Pad R4-1(21.575mm,32.5mm) on Top Layer And Pad R4-2(22.425mm,32.5mm)
Minimum Solder Mask Sliver Constraint (0.047mm < 0.254mm) Between Pad R9-1(24.075mm,105mm) on Top Layer And Pad R9-2(24.925mm,105mm) on
Minimum Solder Mask Sliver Constraint (0.239mm < 0.254mm) Between Via (77.8mm,97.2mm) from Top Layer to Bottom Layer And Via (78.6mm,96mm)

Silk To Solder Mask (Clearance=0.254mm) (IsPad), (All)
Silk To Solder Mask Clearance Constraint (0.224mm < 0.254mm) Between Pad C10-1(115.05mm,35mm) on Top Layer And Text "+" (111.5mm,35.5mm)
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C10-1(115.05mm,35mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C10-1(115.05mm,35mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C10-2(127.55mm,35mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C10-2(127.55mm,35mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.224mm < 0.254mm) Between Pad C18-1(114.25mm,102.5mm) on Top Layer And Text "+" (110.7mm,103mm)
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C18-1(114.25mm,102.5mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C18-1(114.25mm,102.5mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C18-2(126.75mm,102.5mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C18-2(126.75mm,102.5mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C18-2(126.75mm,102.5mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.224mm < 0.254mm) Between Pad C19-1(113.95mm,123.7mm) on Top Layer And Text "+"
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C19-1(113.95mm,123.7mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C19-1(113.95mm,123.7mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C19-2(126.45mm,123.7mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C19-2(126.45mm,123.7mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.224mm < 0.254mm) Between Pad C9-1(115.35mm,12.2mm) on Top Layer And Text "+" (111.8mm,12.7mm)
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C9-1(115.35mm,12.2mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C9-1(115.35mm,12.2mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C9-2(127.85mm,12.2mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.2mm < 0.254mm) Between Pad C9-2(127.85mm,12.2mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R10-1(39.598mm,96.3mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R10-1(39.598mm,96.3mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R10-2(45.802mm,96.3mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R10-2(45.802mm,96.3mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R14-1(82.5mm,83.398mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R14-1(82.5mm,83.398mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R14-2(82.5mm,89.602mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R14-2(82.5mm,89.602mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R2-1(45.298mm,18.3mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R2-1(45.298mm,18.3mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R2-2(51.502mm,18.3mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R2-2(51.502mm,18.3mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R6-1(83.5mm,19.398mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R6-1(83.5mm,19.398mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R6-2(83.5mm,25.602mm) on Top Layer And Trac
Silk To Solder Mask Clearance Constraint (0.204mm < 0.254mm) Between Pad R6-2(83.5mm,25.602mm) on Top Layer And Trac

Silk to Silk (Clearance=0.254mm) (All), (All)
Silk To Silk Clearance Constraint (0.179mm < 0.254mm) Between Text "C8" (45.06mm,9.22mm) on Top Overlay And Trac